UTI Microfluidic Cartridge Diagnostic Report

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Patient: Dr. Vikramaditya Somasekharan Balasubramanian-Mukhopadhyay (72y / Male)
Facility: All India Institute of Medical Sciences (AIIMS) New Delhi — ICMR Advanced AMR Surveillance Reference Lab Node #041
Chief Complaint: Post-transurethral prostatectomy acute severe pyuria, recurrent high fever 39.8°C with rigors, obstructive uropathy

MRN: MRN-AIIMS-DELHI-SURVEILLANCE-NODE-90481-EXTENDED | Specimen: SP-ICMR-MOHFW-2026-999-URINE-CLEAN-CATCH
Clinician: Prof. Dr. Rajeshwar Somasekharan Malhotra, MD, PhD, FAMS (Head of Infectious Diseases)
Collection: Sterile Closed-System Foley Catheter Collection

DIAGNOSTIC STATUS: ACUTE MULTIDRUG-RESISTANT POLYMICROBIAL PYELONEPHRITIS WITH EXTENSIVE RESISTANCE PATTERN AND HIGH RECURRENCE RISK

Severe inflammatory pyuria (>250 WBC/µL) confirmed with dual pathogen co-infection (E. coli and K. pneumoniae). Uropathogen isolates exhibit widespread fluoroquinolone resistance and beta-lactam resistance. [HIGH-RISK RECURRENCE PATTERN: Patient has 14 prior documented episodes over 12 months with indwelling catheterization. Cross-resistance emerging across second-line agents. Immediate parenteral therapy with Carbapenems or culture-guided IV aminoglycosides warranted per hospital antimicrobial stewardship policy. Ensure renal function monitoring.]

1. Pyuria / Pus-Cell Assay (Layer 1 — Leukocyte Esterase)

INFECTION SEVERITY	PYURIA WBC ESTIMATE	SIGNAL OPTICAL DENSITY
HIGH	> 250 WBC/µL (Severe Acute Inflammatory Pyuria)	0.842 OD

2. 4-Plex Pathogen Aptamer Panel (Layer 2 — Aptamer Capture)

TARGET ORGANISM	GRAM STAIN	OPTICAL DENSITY	STATUS
Escherichia coli (UPEC clone ST131)	**Gram-negative bacillus**	**0.720 (Cutoff: 0.22)**	**[DETECTED]**
Klebsiella pneumoniae subsp. pneumoniae	**Gram-negative encapsulated bacillus**	**0.650 (Cutoff: 0.2)**	**[DETECTED]**
Proteus mirabilis (Urease Producer)	Gram-negative swarming bacillus	0.040 (Cutoff: 0.2)	Not Detected
Pseudomonas aeruginosa (Mucoid strain)	Gram-negative non-fermenter	0.040 (Cutoff: 0.21)	Not Detected

3. Antibiotic Susceptibility Testing (Layer 3 — 4× Replicate Panel)

ANTIMICROBIAL AGENT	CLASS / LINE	MEAN OD ± SD	CONCORDANCE	RESULT
Ampicillin	Aminopenicillin (Cell Wall Inhibitor)	0.790 ± 0.04	4/4 concordant (100%)	**RESISTANT**
Ciprofloxacin	Fluoroquinolone (DNA Gyrase Inhibitor)	0.812 ± 0.03	4/4 concordant (100%)	**RESISTANT**
Nitrofurantoin* (1st-Line)	Nitrofuran (Bacterial Ribosome Inactivator)	0.310 ± 0.05	2/4 concordant (50%)	**INTERMEDIATE**
Cotrimoxazole* (1st-Line)	Folate Synthesis Inhibitor (DHFR/DHPS)	0.740 ± 0.02	4/4 concordant (100%)	**RESISTANT**
Fosfomycin* (1st-Line)	Phosphonic Acid Derivative (MurA Inactivator)	0.062 ± 0.01	4/4 concordant (100%)	**SENSITIVE**

ANTIMICROBIAL STEWARDSHIP & FIRST-LINE RECOMMENDATIONS:
Sensitive First-Line Options: Fosfomycin. Reserve fluoroquinolones (Ciprofloxacin) to preserve efficacy and prevent resistance escalation.

LEGAL MEDICAL DISCLAIMER: Genex is an investigational point-of-care software reader for microfluidic cartridges. This report does not constitute a certified medical diagnosis and is intended to support clinical decision-making. Treatment must be confirmed by a licensed medical practitioner in accordance with local antimicrobial guidelines.